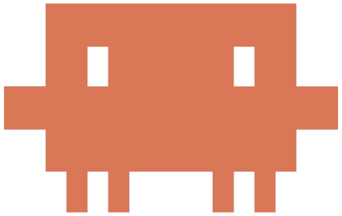
Acceleration and variability in vocabulary growth for young children (and language models)

Michael C. Frank

AI disclosure

- First project I’ve collaborated completely with AI (Claude Code)
- ~6 weeks of constant iteration
- Being careful about requesting cleanup, documentation, best practices throughout
- Question about how to establish trust?
  - Key plots to diagnose model performance
  - Annotated Stan code
  - Provenance and reproducibility traces for all figures/table



Two views of early word learning

Slow and incremental

- Statistical learning (Saffran et al., 1996)
- Cross-situational learning (Yu & Smith, 2007)
- Input-uptake relationships (Weisleder & Fernald, 2012)

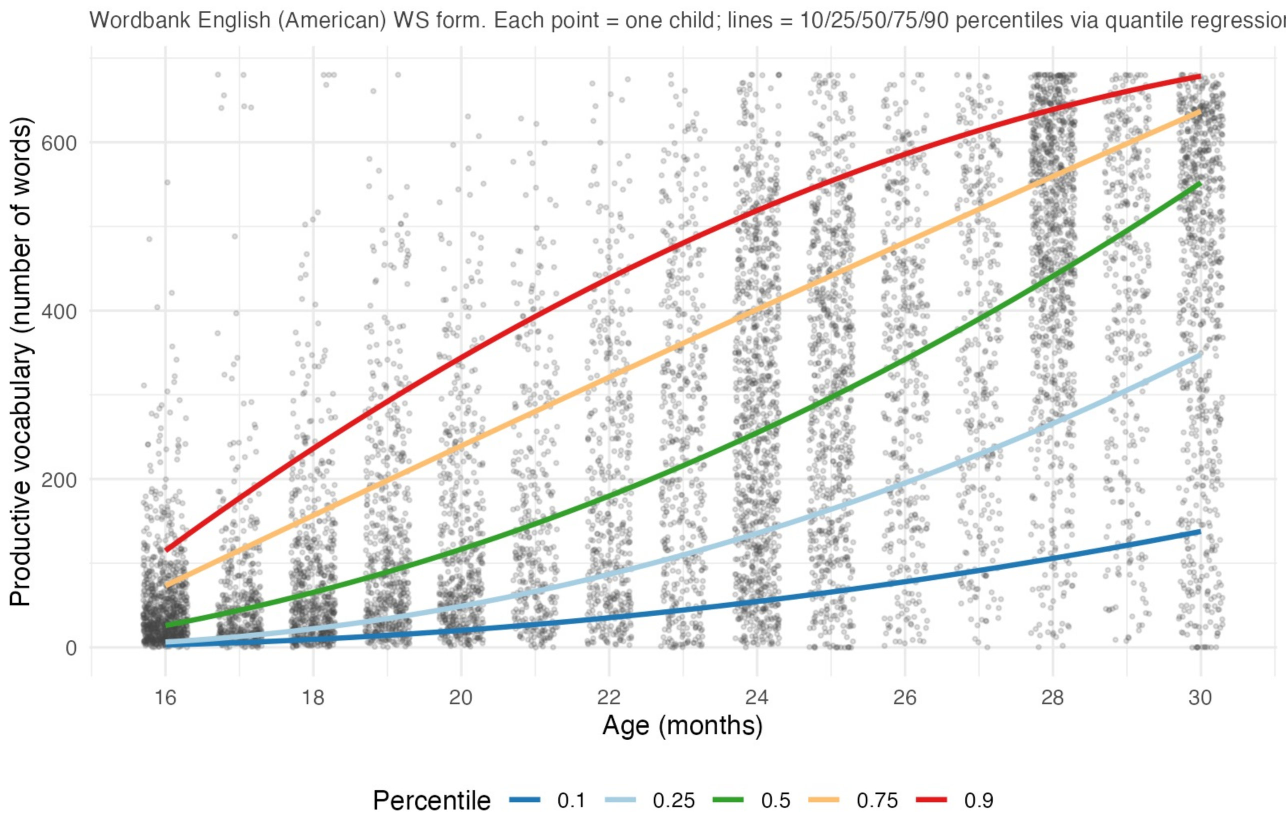
Quick and inferential

- Mutual exclusivity (Markman et al., 1988)
- Shape bias (Smith et al., 2002)
- Pragmatic word learning (Tomasello, 2000)

Exploring signatures of vocabulary growth

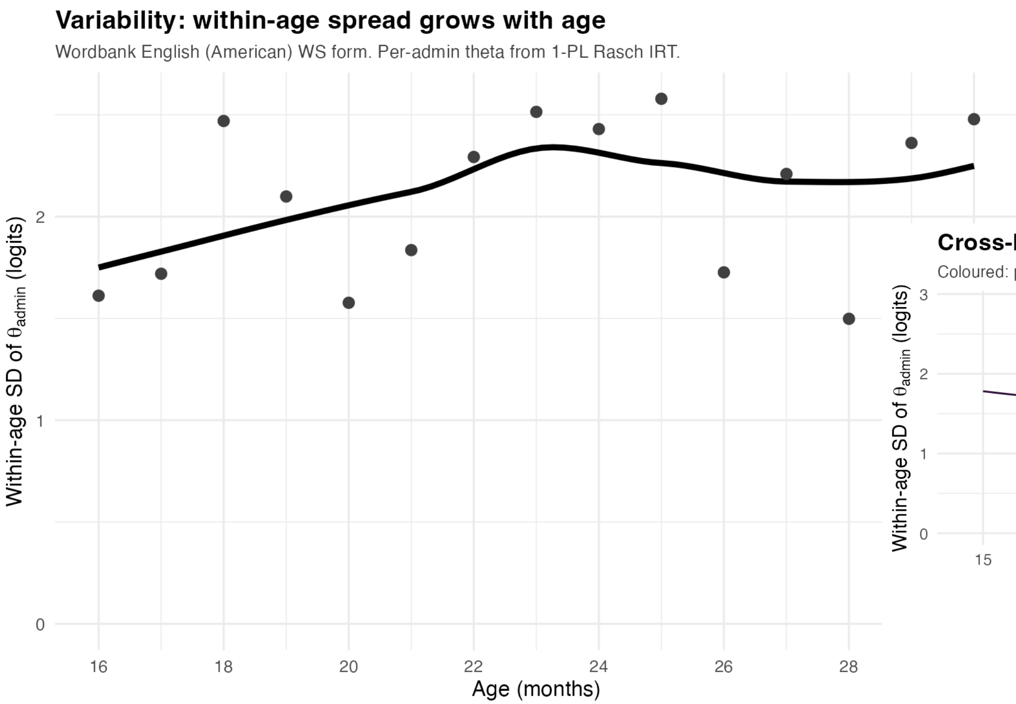
Acceleration + variability are **ubiquitous**

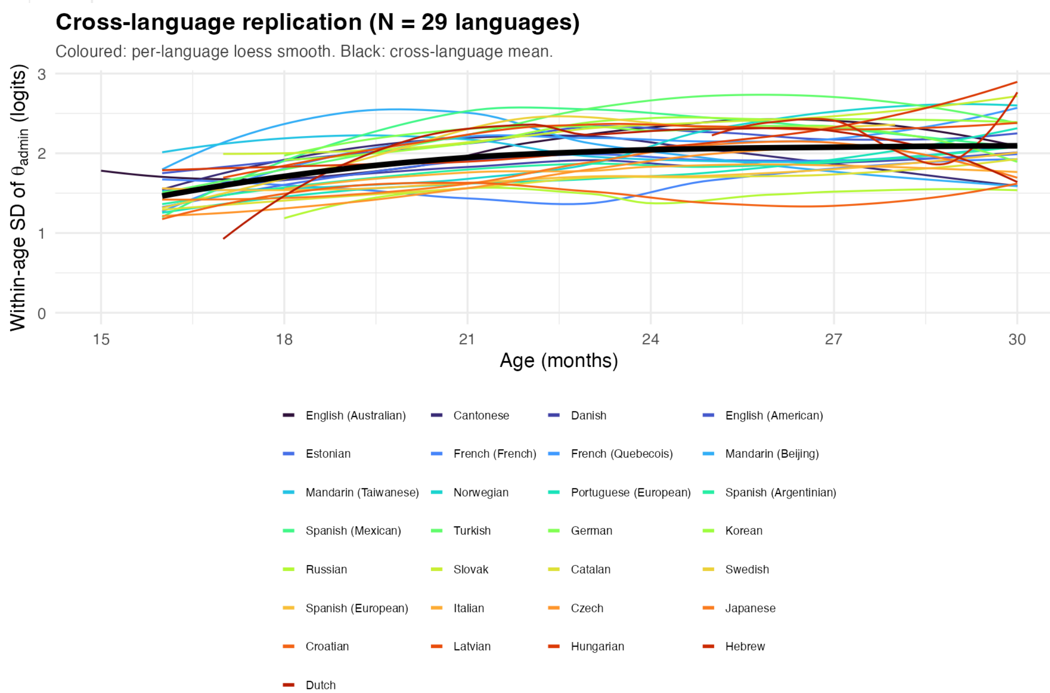
Can they help us reconcile these two views?



Terminology note: **Efficiency:** what's changing in the child**; Acceleration:** what's happening in the data

Two signatures of vocabulary growth





Picking up where George left off...

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Toward a "Standard Model" of Early Language Learning

George Kachergis, Virginia A. Marchman, and Michael C. Frank
Department of Psychology, Stanford University

Abstract

A standard model is a theoretical framework that synthesizes observables into a quantitative consensus. Have researchers made progress toward this kind of synthesis for children's early language learning? Many computational models of early vocabulary learning assume that individual words are learned through an accumulation of environmental input. This assumption is also implicit in empirical work that emphasizes links between language input and learning outcomes. However, models have typically focused on average performance, whereas empirical work has focused on variability. To model individual variability, we relate the tradition of research on accumulator models to item response theory models from psychometrics. This formal connection reveals that currently available data sets do not allow researchers to test the resulting models fully, illustrating a critical need for theory to contribute to shaping new data collection and creating and testing an eventual standard model.

Is growth pure accumulation?

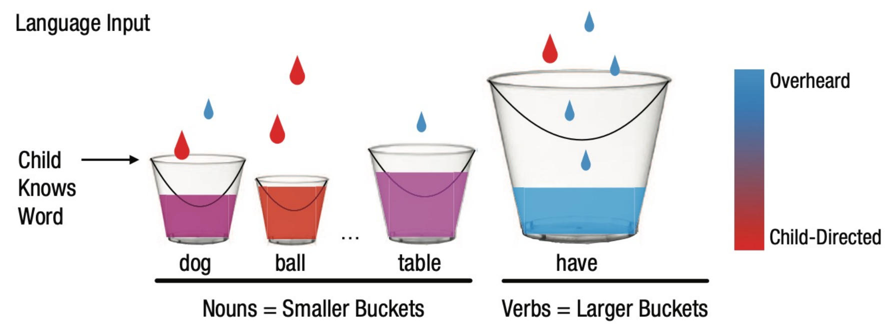
Maybe acceleration just falls out of accumulation?

Each child has input rate r_i; word j has difficulty δ_j. A word is acquired when enough tokens accumulate.

Derived from cross-situational word learning and other associative/statistical learning ideas.

Acceleration: Some models have changes in efficiency to create acceleration, others do not.

Variation: Between-child variation only in baseline (r_i or α_i). No variation in efficiency.



In the literature:

- McMurray (2007) — pure rate × time
- Mitchell & McMurray (2009) — + leverage feedback
- Hidaka (2013) — cumulative-Weibull; rate change
- Mollica & Piantadosi (2017) — per-word Gamma
- **Kachergis et al. (2021) — IRT Rasch + α_i** (we build on this model)

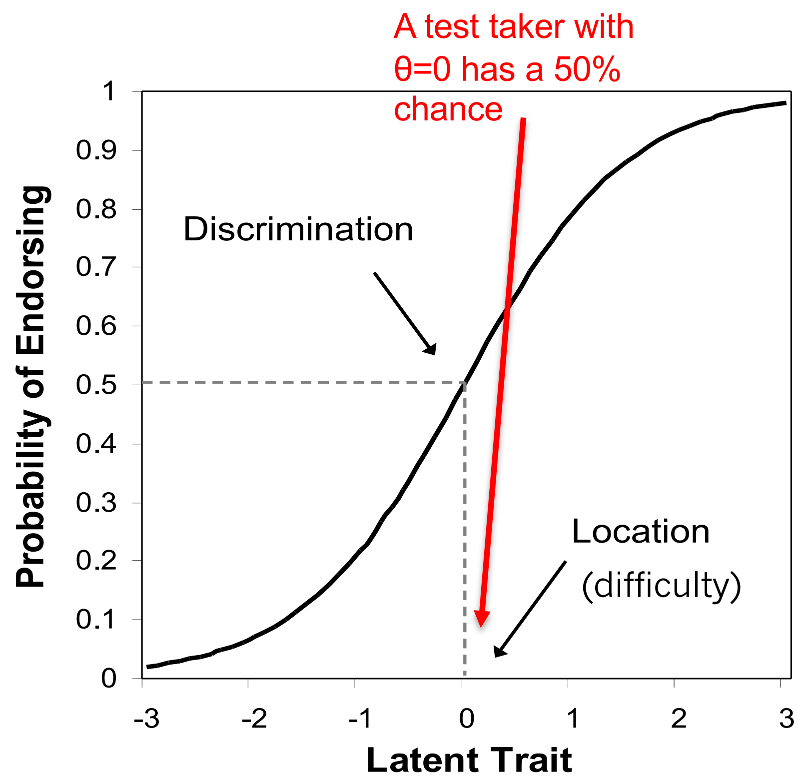
Item response theory (IRT)

IRT describes a family of models of binary data (like CDI data).

We use the Rasch model, a logistic model where each item has a single difficulty parameter δ, and each person has a single ability parameter θ.

Probability of success on an item is then just:

$$P(w_{i,j} = 1 | \theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1 + exp(\theta_i - \delta_j)}$$



ICC = item characteristic curve

Puzzle: input matters less than supposed

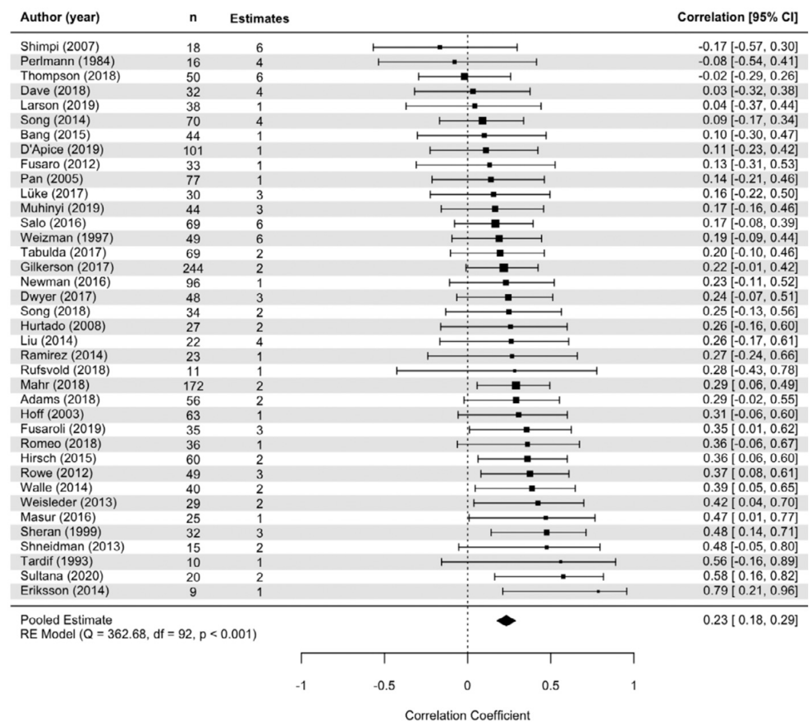
The accumulator view predicts variation in growth comes from variation in input quantity.

Coffey & Snedeker (2026) meta-analyzed 71 studies, 4,760 children:

R^2 of input on outcome = 0.04–0.07.

Caregiver speech quantity explains 4–7% of variance in language outcome — a real but small effect.

Hypothesis: Variability comes from *how children accelerate*, not how much input they hear.



This work: modeling individual variability

We build a Bayesian model based on Kachergis et al. (2021) “standard model” to estimate:

1. **Efficiency change.** Does acceleration result from changes in learning efficiency?
2. **Input share.** How much of the between-child variance is attributable to input rate vs. to internal factors like variation in efficiency?

Does the same machinery applied to large language models yield the same answers?

BUILDING THE MODEL

The accumulator hypothesis

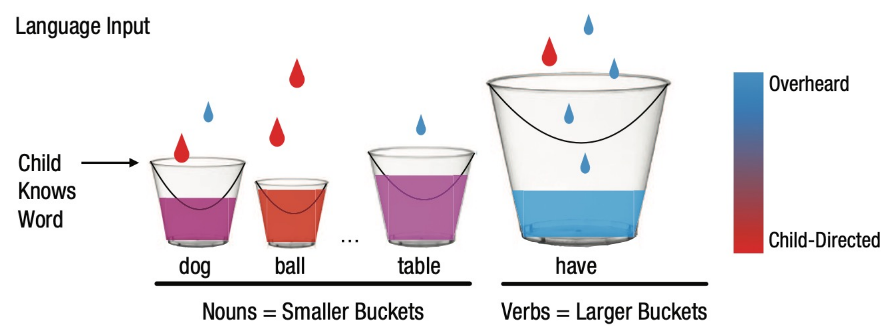
The idea. Each child accumulates word-evidence at her own input rate r_i; word j is acquired once its difficulty δ_j is crossed.

Ability scales as a power of input. If kids vary only in *rate*, then ability grows as $(\text{tokens})^1$ — the pure *unit accumulator*.

Between-child differences come only from differences in r_i.

Model checks:

- Is the growth exponent really 1, or steeper?
- Are childrenalike except for input rate, or do they also differ in efficiency (intercept) or in efficiency change (slope)?



Kachergis et al. (2021)

Rasch model

$P\left(w_{i,j} = 1 | \theta_i, \delta_j\right) = \frac{\exp(\theta_i - \delta_j)}{1 + exp(\theta_i - \delta_j)}$:

$$\underbrace{\theta_i}_{\substack{\textbf{kid ability} \\ \textit{(grows with experience)}}} \quad - \quad \underbrace{\delta_j}_{\substack{\textbf{word difficulty} \\ \textit{(per-word, free)}}}$$

Standard Rasch IRT model – the substantive content is in how θ_i depends on accumulated experience.

Rasch model

$$P\left(w_{i,j} = 1 | \theta_i, \delta_j\right) = \frac{\exp(\theta_i - \delta_j)}{1 + exp(\theta_i - \delta_j)}:$$

$$\underbrace{\theta_i}_{\textbf{kid ability}} \quad - \quad \underbrace{\delta_j}_{\textbf{word difficulty}}$$

kid ability
(grows with experience)

word difficulty
(per-word, free)

Focus of most prior work (e.g., Kachergis et al., 2021; various AOA prediction work) – we leave this free here and don't predict individual word difficulties

Rasch + accumulator dynamics

$P(w_{i,j} = 1|\theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1+exp(\theta_i - \delta_j)}$, where:

Different from Kachergis; matches Peekbank & Hidaka (2013). Power law, not exponential!

$$\theta_i = \underbrace{\log r_i}_{\substack{\text{per-child input rate} \\ \text{(words per month)}}} + \underbrace{\log t}_{\substack{\text{time = age} \\ \text{(months since onset)}}}$$

Ability = rate × time. The only thing that changes across development is more accumulated input. The only source of variation between children is variation in the input rate

Rasch accumulator + per-child efficiency

$P\left(w_{i,j} = 1 | \theta_i, \delta_j\right) = \frac{\exp(\theta_i - \delta_j)}{1 + exp(\theta_i - \delta_j)}$, where:

$$\theta_i = \underset{\substack{\uparrow \\ \textit{input rate}}}{\log r_i} + \underset{\substack{\uparrow \\ \textit{per-child} \\ \textit{efficiency} \\ \textit{(NEW)}}}{\log \alpha_i} + \underset{\substack{\uparrow \\ \textit{time}}}{\log t}$$

Each child has a multiplicative efficiency α_i over and above their input rate.

How much of the between-child variance comes from efficiency vs. input rate? We summarize this share as $\pi_\alpha = \sigma_\alpha^2 / (\sigma_\alpha^2 + \sigma_r^2)$.

Rasch accumulator + efficiency slope

$P(w_{i,j} = 1|\theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1+exp(\theta_i - \delta_j)}$, where:

$$\theta_i = \underbrace{\log r_i}_{\textit{input rate}} + \underbrace{\log \alpha_i}_{\textit{efficiency}} + \underbrace{\kappa_{\text{pop}}}_{\textit{population scaling exponent (NEW; shared)}} \cdot \underbrace{\log t}_{\textit{time}}$$

The population accelerates: ability scales as $t^{\kappa_{\text{pop}}}$ super-linearly. All children still share one κ_{pop}.

Is κ_{pop} really above 1 — i.e., does ability grow faster than a unit accumulator predicts? This is Hidaka's rate-change model (same as Kachergis et al., 2021).

Rasch accumulator + per-child efficiency slope

$P\left(w_{i,j} = 1 | \theta_i, \delta_j\right) = \frac{\exp(\theta_i - \delta_j)}{1 + exp(\theta_i - \delta_j)}$, where:

$$\theta_i = \log r_i + \underbrace{\log \alpha_i}_{\textit{efficiency}} + \Big(\underbrace{\kappa_{\text{pop}}}_{\textit{pop. scaling}} + \underbrace{\zeta_i}_{\substack{\textbf{\textit{per-child}} \\ \textbf{\textit{acceleration}} \\ \textbf{\textit{deviation}} \\ \textbf{\textit{(NEW)}}}} \Big) \cdot \log t$$

Each child has their own growth exponent $\kappa_i = \kappa_{\text{pop}} + \zeta_i$. The new piece: between-child variation in how efficiency changes.

Rasch – glmer equivalency

This model is also a GLMER with:

```
produces ~ = 1 + log_age
  + (1 + log_age | child)
  + (1 | word)
```

modulo some significant (but not huge) differences…

- Can pin input rate externally
- Bayesian/extensible
- (We never examined fit to variance in prior work!)

RESULTS: ENGLISH AND NORWEGIAN LONGITUDINAL FITS

Data and compute for the longitudinal fits

Data.

- **Reduced English (WS form).** I = 200 children, J = 200 words.
- **English.** $I = 500$ children, $\sim$ 1500 administrations, $J = 671$ items (full WS list), $N \approx 1.1$M observed responses. Stratified-by-median-age sampling.
- **Norwegian.** $I = 500$ children, $\sim$ 3000 administrations, $J = 716$ items, $N \approx 2.2$M responses. Same stratification.

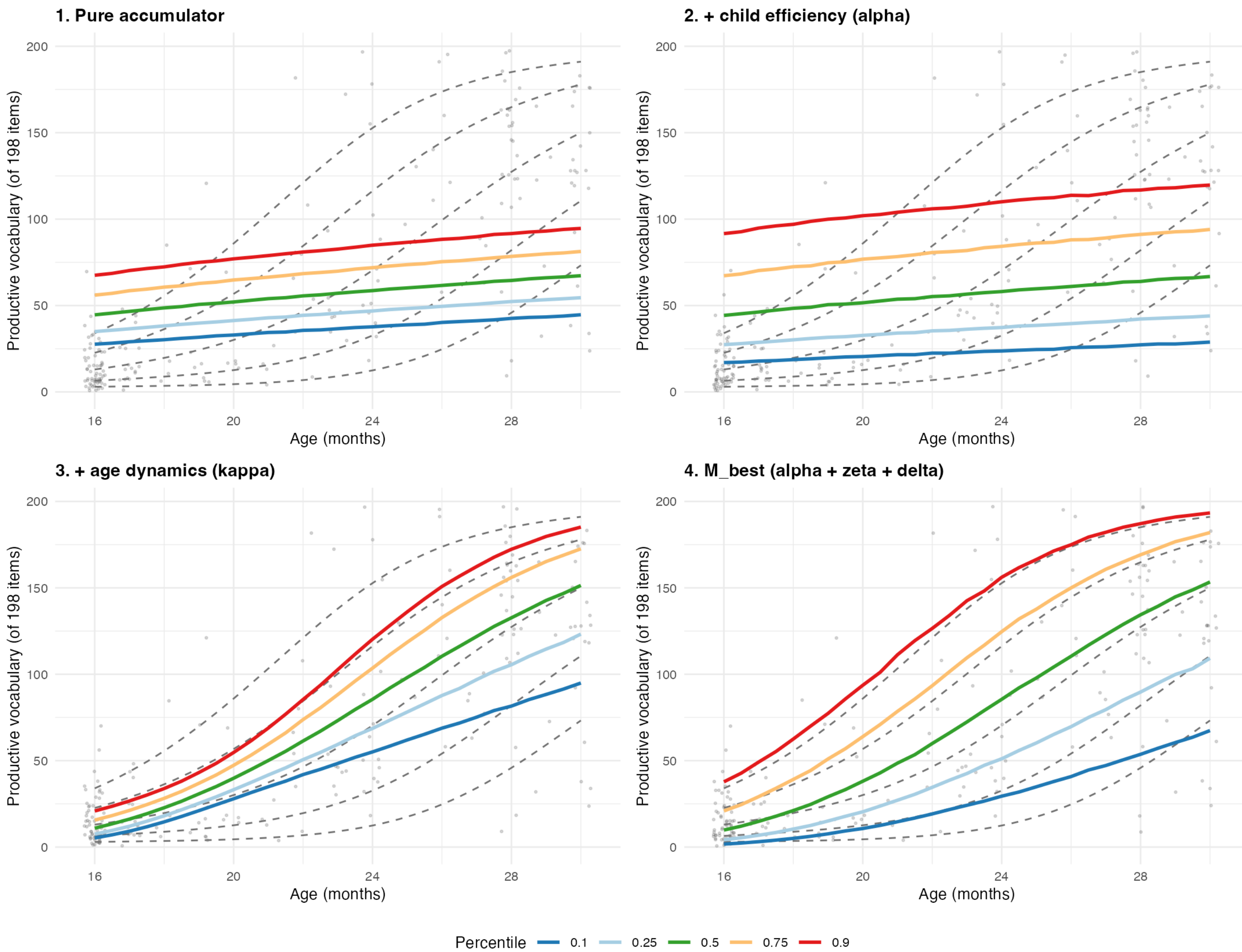
Model family. Four-step build run as a sequence of Stan models on each dataset: `pure`, $+\,\alpha$, $+\kappa_{\text{pop}}$, $+\zeta_i$.

Compute.

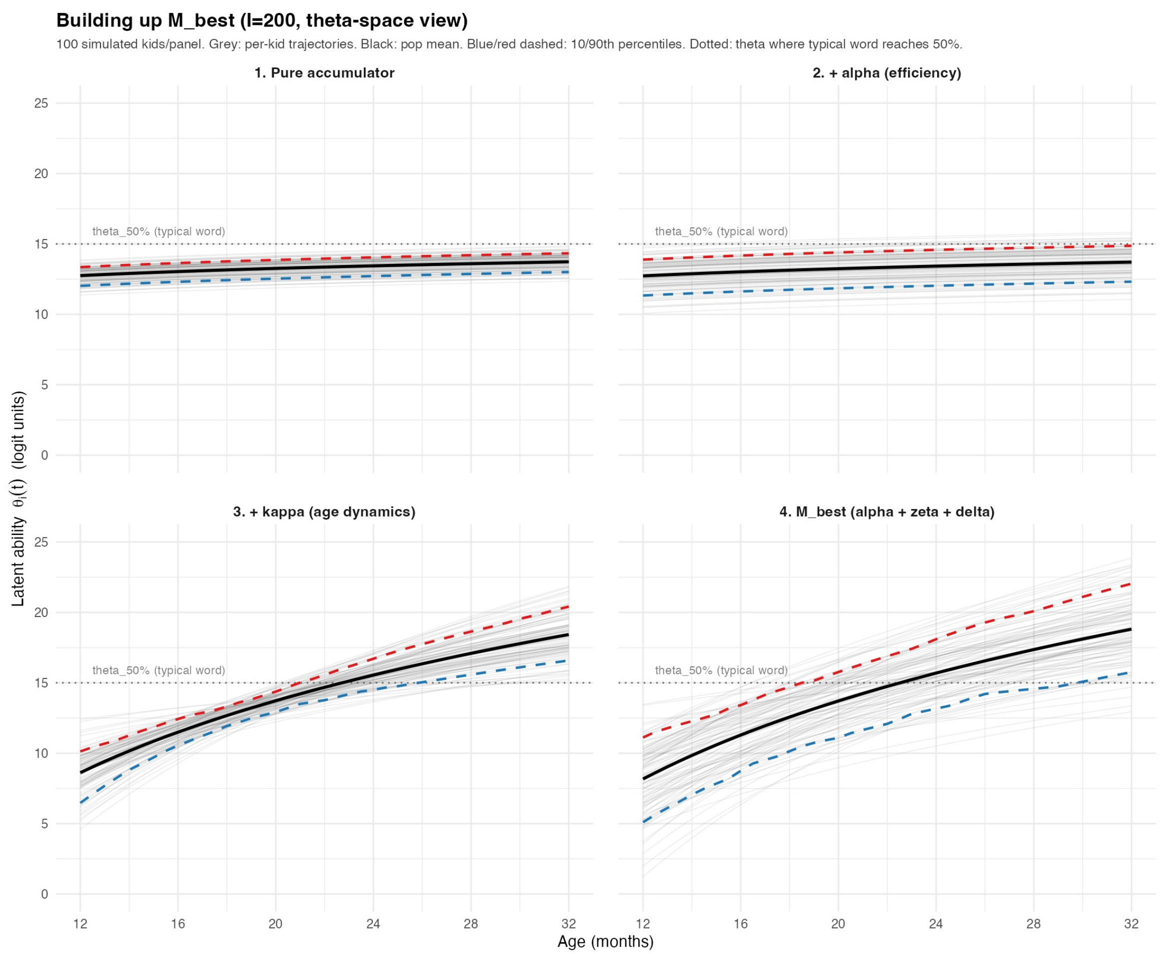
- Sherlock for reduced fits (all CPU).
- GCP virtual machine (`n2d-standard-128`, 64 physical AMD EPYC cores each, 128 GB RAM). Stan with `reduce_sum` threading: 4 chains × 16 threads/chain saturates the 64 physical cores per machine.
- Check by fitting glmer 🙃

Building up M_best (English I=200, J=198)

Each panel = one component added. Empirical: grey dots (1 admin/kid x-sec) + dashed lines (GAMLSS BEINF lambda=10000).

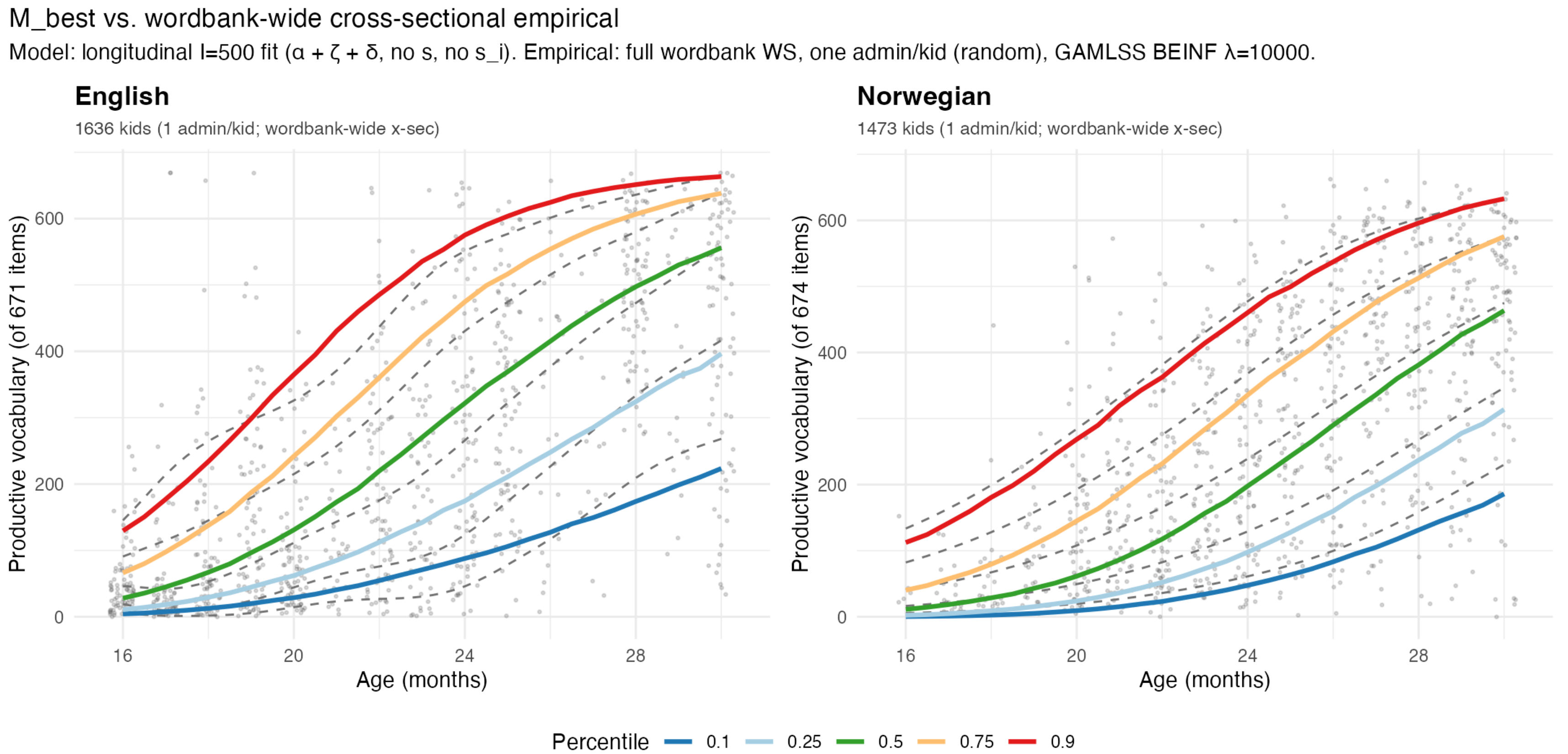


Latent ability changes



Note power law not exponential – hard to think about – fastest acceleration early on, not later

N=500 longitudinal data



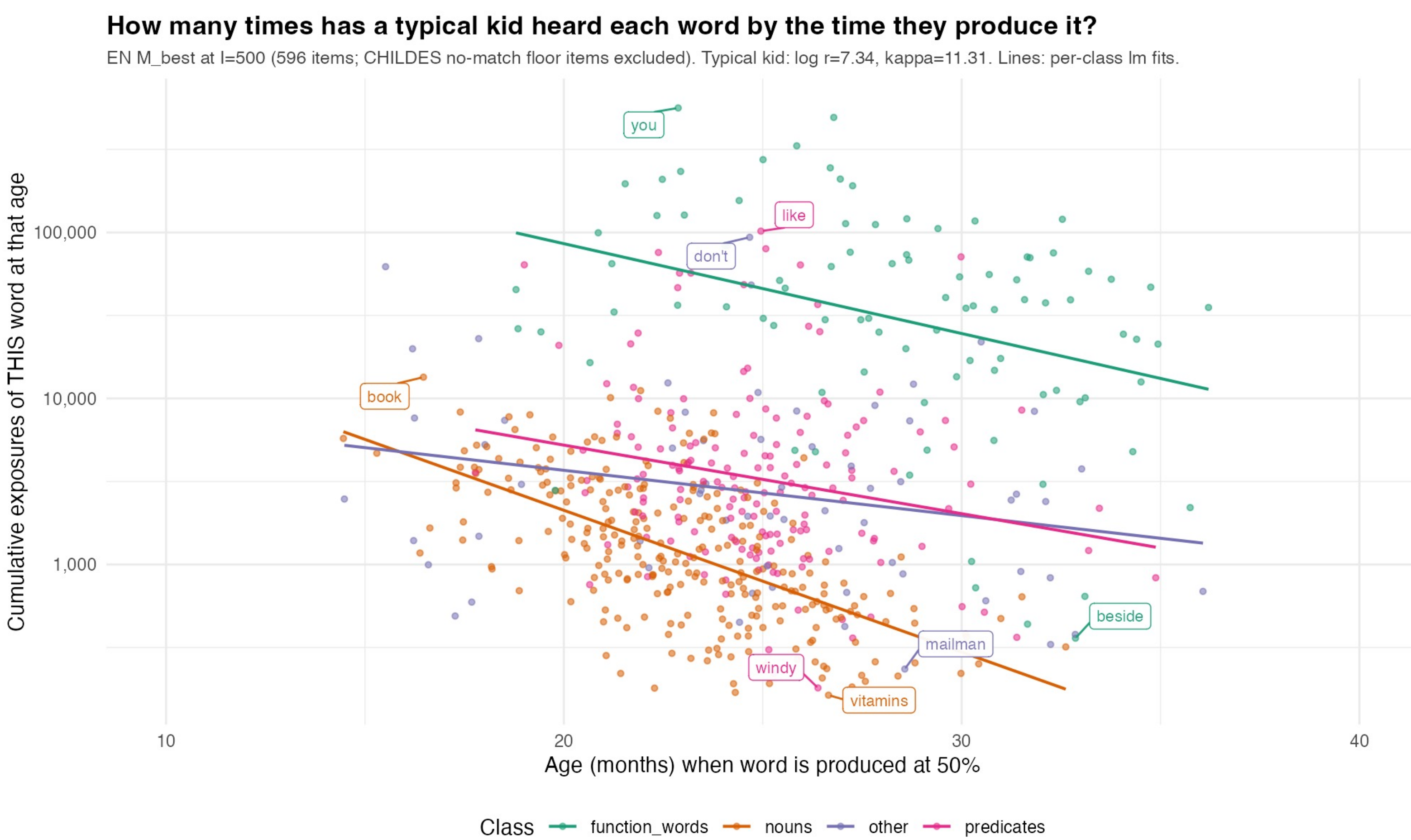
Best-fitting model: parameter posteriors

Parameter	EN M_best (I=500)	NO M_best (I=500)
I (kids)	500	500
J (items)	671	674
N (obs)	1106805	2056872
kappa	10.31 [10.25, 10.37]	11.47 [11.40, 11.53]
sigma_alpha	1.81 [1.69, 1.94]	2.05 [1.93, 2.18]
sigma_zeta	3.81 [3.58, 4.07]	4.79 [4.49, 5.13]
pi_alpha	0.92 [0.91, 0.93]	0.94 [0.93, 0.94]
rho(xi, zeta)	-0.09 [-0.18, -0.01]	-0.13 [-0.21, -0.04]

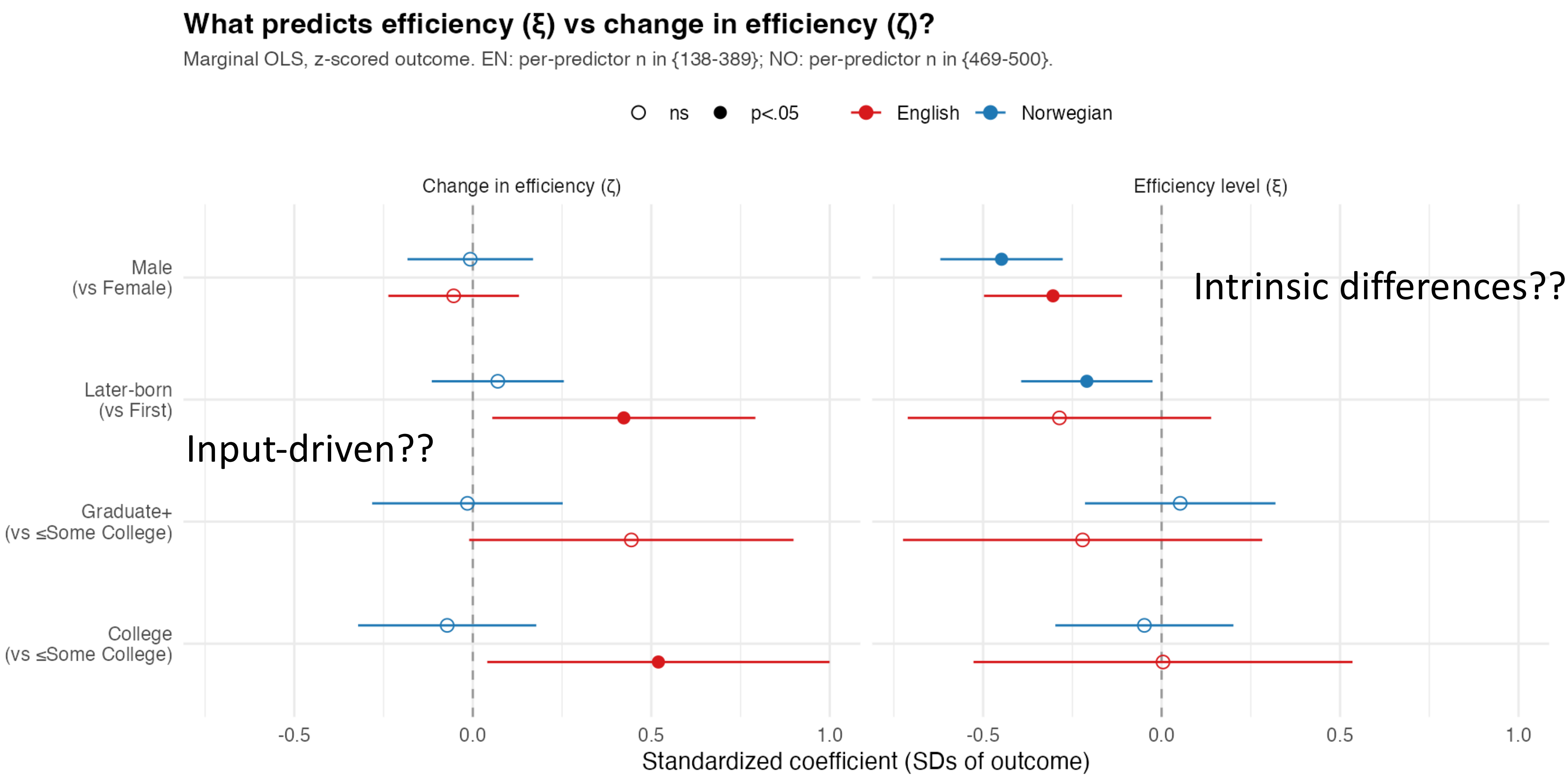
Increasing efficiency!

Efficiency variation >> rate variation

Exponentially fewer exposures before learning



Demographic variation across two parameters



INPUT QUANTITY

Where does σ_r come from? The input-rate prior

The kid-level intercept $\xi_i = \log r_i + \log \alpha_i$ has total variance σ_ξ^2 that the data *does* identify. But the split between input rate (σ_r^2) and efficiency (σ_α^2) is not separately identified within the model:

$$\sigma_\xi^2 \;=\; \sigma_r^2 + \sigma_\alpha^2 \Rightarrow \pi_\alpha \;=\; 1 - \frac{\sigma_r^2}{\sigma_\xi^2}$$

We resolve this by pinning σ_r externally — from data quantifying actual variation in child-directed input rates across families. The model then partitions the data-identified σ_ξ^2 into input and efficiency components.

The pin matters for π_α.

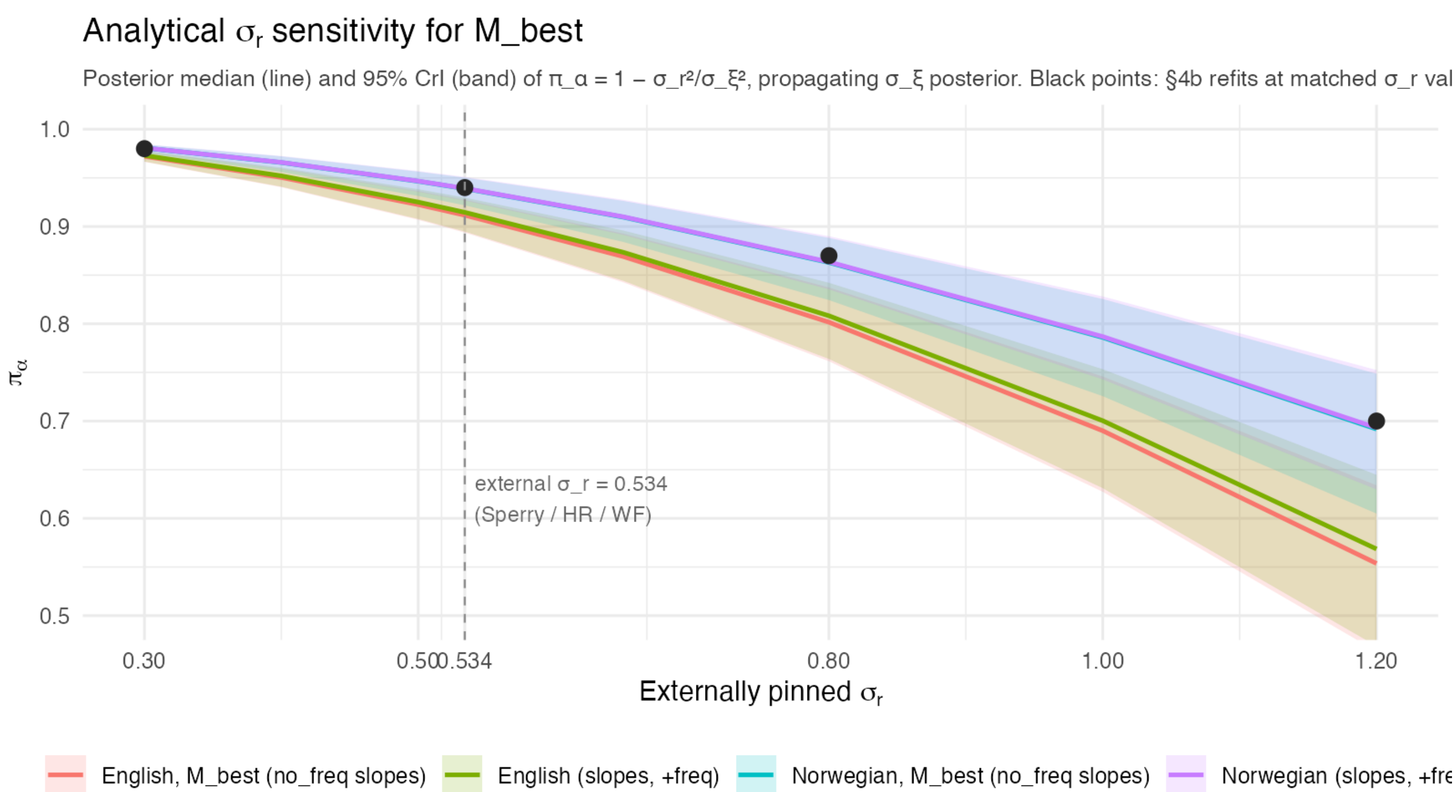
- Bigger $\sigma_r \Rightarrow$ more variance assigned to input $\Rightarrow$ lower π_α.
- Pinning σ_r *too low* would overstate efficiency's share; *too high* would mask it.

Strategy. Compile the empirical literature on adult-token rates per child, derive a defensible σ_r, and report a full sensitivity sweep so the headline doesn't hinge on the central pin.

Estimating input variation

Source	Sample	Channel	n	tok/hr (mean)	tok/hr (±1 SD)	tok/mo (mean)	tok/mo (±1 SD)	σ_r (log)	1-SD factor
Hart & Risley	all pooled	mother CDS	42	1,148	546 – 2,410	419K	199K – 880K	0.74	2.10×
Hart & Risley	within-stratum	mother CDS	42	1,148	631 – 2,086	419K	230K – 761K	0.6	1.82×
Soderstrom & Wittebolle	all pooled	all speech	12	967	622 – 1,502	353K	227K – 548K	0.44	1.55×
Sperry (PIN)	**all pooled**	**adult-to-child**	**42**	**1,537**	**901 – 2,622**	**561K**	**329K – 957K**	**0.53**	**1.71×**
Sperry	within-stratum	adult-to-child	42	1,537	974 – 2,425	561K	356K – 885K	0.46	1.58×
Weisleder & Fernald	all pooled	mother CDS	29	458	197 – 1,065	167K	72K – 389K	0.84	2.33×
Bergelson (SEEDLingS LENA)*	US 13–18 mo daylong	LENA AWC	44	1,100	650 – 1,850	400K	240K – 675K	0.52	1.68×
Casillas (Tseltal Mayan)*	Chiapas daylong	adult-to-child	10	500	170 – 1,500	180K	62K – 550K	1.09	2.97×
Bunce et al. 2024 (meta)*	11 communities	adult-to-child	82	900	280 – 2,900	330K	100K – 1,060K	1.18	3.25×

Sensitivity: π_α across plausible σ_r



Central pin $\sigma_r = 0.534$**:** $\pi_\alpha \approx 0.91 - 0.94.$

Doubled $\sigma_r = 1.2$ **(cross-cultural upper bound):** π_α still $> 0.55.$

Halved $\sigma_r = 0.30$ **(below any defensible single-site SD):** $\pi_\alpha \approx 0.97.$

Caveat: input *quantity*, not *quality*

Haven't modeled variation in input quality — diversity, contingency, conversational structure, child-directed register. Two distinct effects:

1. Constant per-child quality — refines what σ_α is
Fixed multiplier on input: q_i constant in age - indistinguishable from σ_α. So part of the σ_α we report could be hidden quality variance:

target π_α	σ_q needed	per-SD factor on quality
0.90	0.27	1.31×
0.70	0.88	2.42×
0.50	1.22	3.40×

For π_α to drop below 0.5, kid-to-kid quality variation would need $\sigma_q \approx 1.22$ — more than 2× as much as measured input-rate variation ($\sigma_r \approx 0.53$).

2. Age-varying quality — competes with κ (acceleration)
A quality multiplier that grows with age is equivalent to a steeper κ_i. To absorb the observed acceleration into a quality story, "quality" would have to grow ~3,000× for the median kid,

Quality might explain part of σ_α but cannot explain acceleration.

EXTENSIONS: DIRECTLY OBSERVED INPUT AND PROCESSING

Model extensions: testing π_α with richer data

Wordbank longitudinal fits use *population-level* input rate priors (Sperry/Hart-Risley/Weisleder-Fernald).

Two natural follow-ups:

1. **Per-child observed input**: use *measured* per-child input rate instead of a population prior. Does π_α stay ~ 0.9 when input is no longer a pooled estimate? → **BabyView, SEEDLingS.**

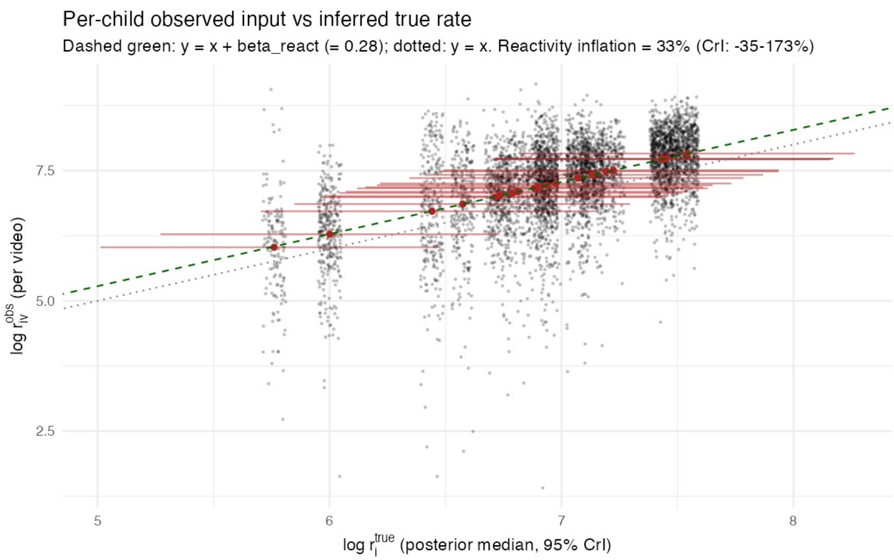
2. **Per-child processing efficiency**: link CDI growth to an independent per-child readout of processing speed. Does the σ_α we estimate from CDI reflect "processing efficiency" or something else? → **Peekbank.**

Both extensions reuse the same model machinery with an added channel in the linear predictor; only the data and the per-child latents change.

Replicating with directly observed input

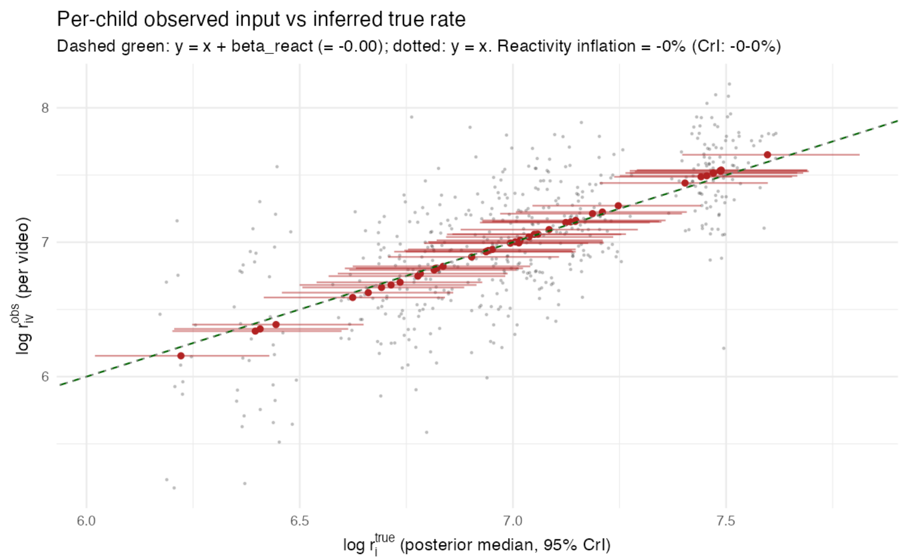
BabyView, $N = 20$

$\pi_\alpha = 0.84$, [0.68, 0.93]

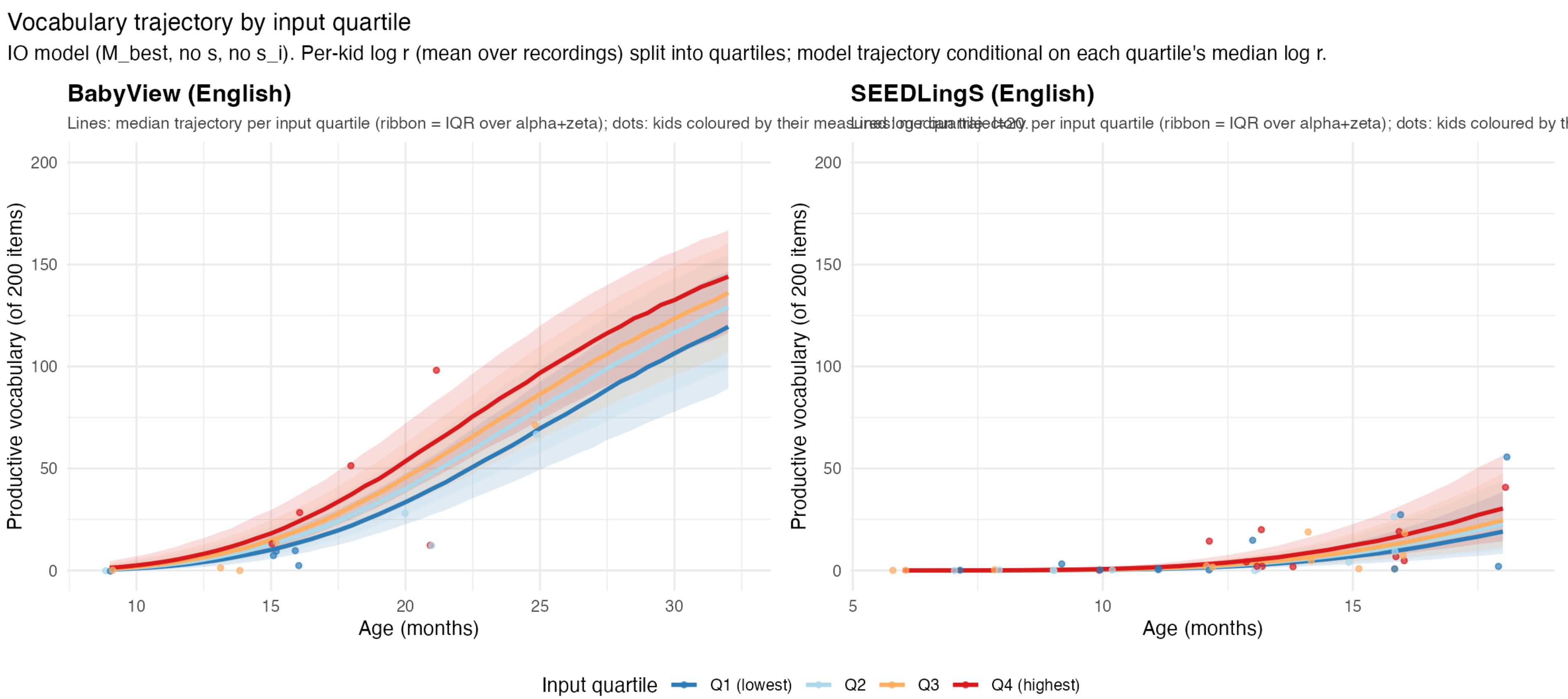


SEEDLingS, $N = 44$ **(comp corrected)**

$\pi_\alpha = 0.94$, [0.89, 0.97]



Minimal input effects



Processing speed readout: Peekbank LWL

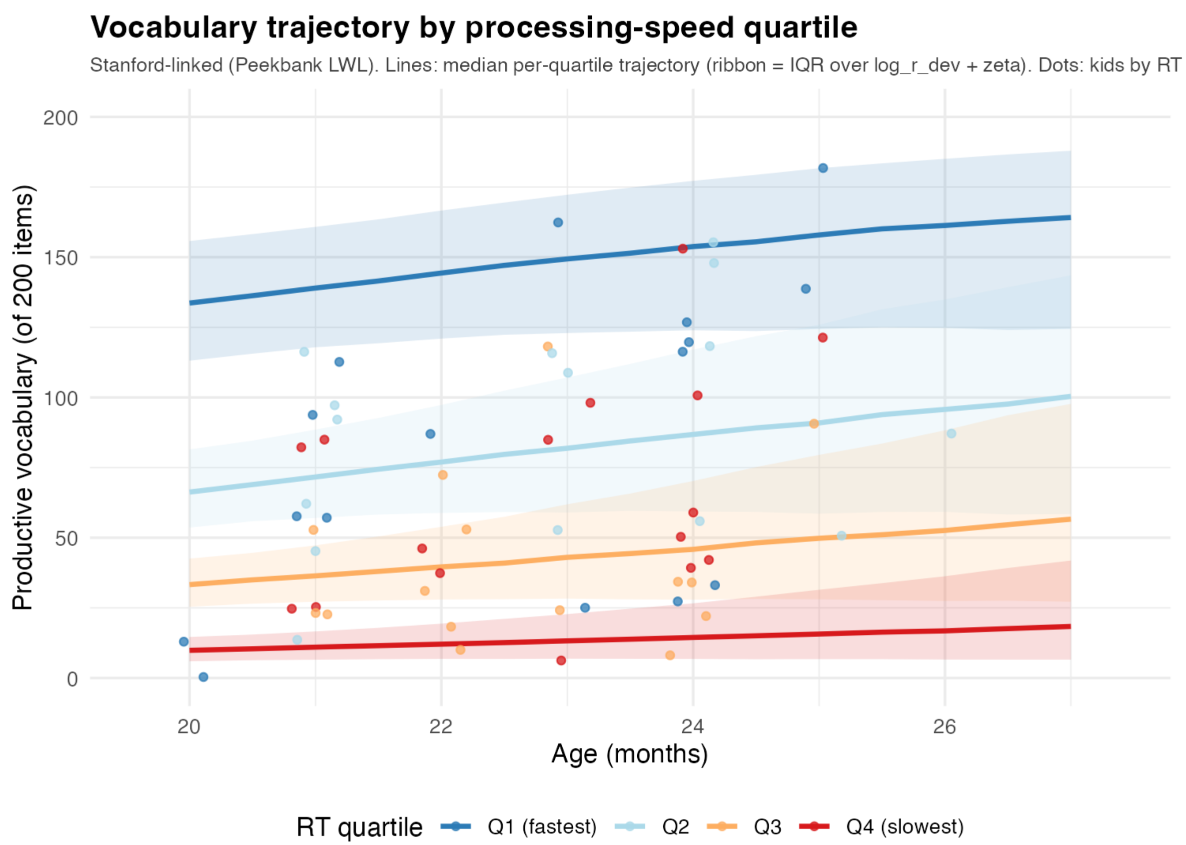
Joint model adds a per-child LWL log-RT readout coupled to $\log \alpha_i$ via γ_{rt}.

- γ_rt = 0.084, [0.044, 0.125]
  - vocab efficiency *level* (log α) and processing-speed *level* (RT intercept) are coupled: ~13% RT speedup per SD of α.
- π_α = 0.90, [0.86, 0.94] — replicates within Peekbank.
- ρ(ζ, rtslope) = +0.05, [–0.26, +0.34]
  - but vocab acceleration and RT *maturation* are uncoupled.

Shared efficiency (intercept), distinct efficiency change (slopes)

Replicates Peekbank paper (Frank et al., 2026, *eLife*): higher RT = bigger vocab per age, but not faster *growth*

Peekbank RT results



π_α across five samples

Parameter	EN M_best (I=500)	NO M_best (I=500)	BabyView (IO)	SEEDLingS (IO)	Peekbank (proc)
I (kids)	500	500	20	44	62
J (items)	671	674	200	200	200
N (obs)	1106805	2056872	16459	102800	20600
kappa	10.31 [10.25, 10.37]	11.47 [11.40, 11.53]	6.42 [5.98, 6.87]	8.19 [7.82, 8.57]	2.51 [1.60, 3.38]
sigma_alpha	1.81 [1.69, 1.94]	2.05 [1.93, 2.18]	1.13 [0.82, 1.56]	1.39 [1.12, 1.74]	1.67 [1.35, 2.03]
sigma_zeta	3.81 [3.58, 4.07]	4.79 [4.49, 5.13]	2.57 [1.92, 3.39]	3.97 [3.32, 4.70]	6.42 [5.34, 7.54]
pi_alpha	0.92 [0.91, 0.93]	0.94 [0.93, 0.94]	0.83 [0.66, 0.93]	0.94 [0.89, 0.97]	0.90 [0.86, 0.94]

CONNECTING TO LLMS

Why look at LLMs?

LLMs and children both learn word meanings from input. They differ in scale and architecture, but the question of how their learning scales with input is statistically well-posed:

- For an LM and a word w, we have $P(w$ produced correctly) as a function of training tokens.
- This is a sigmoid in log-tokens with a per-word slope.
- The same statistic exists for human children: $P(w$ acquired) as a function of cumulative input.

This lets us put kids and LMs on the same axis:

1. Do kids and LMs have similar per-word acquisition slopes?
2. Does a model that fits kids' learning say anything about LMs?

Chang & Bergen: Per-word sigmoids in LMs

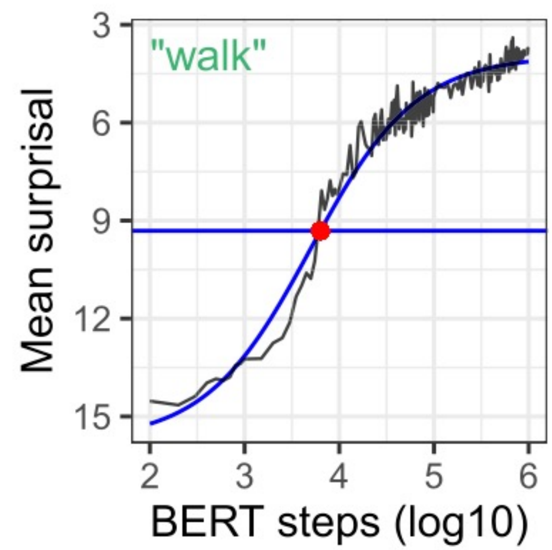
Setup. Train LM (BERT, biLSTM, GPT-2, LSTM). At each training checkpoint, probe each CDI word to get surprisal.

For each (LM, word), fit a 4-parameter logistic

$$S_w(x) = \mathrm{lower}_w + \frac{\mathrm{upper}_w - \mathrm{lower}_w}{1 + \exp\big((x - \mathrm{mid}_w)/\mathrm{scale}_w\big)}$$

where $x = \log_{10}$(training steps), yielding a per-word **scaling slope**.

After unit conversion, this is exactly analogous to our $\kappa_i \log t$ for children.



Chang & Bergen (2022)

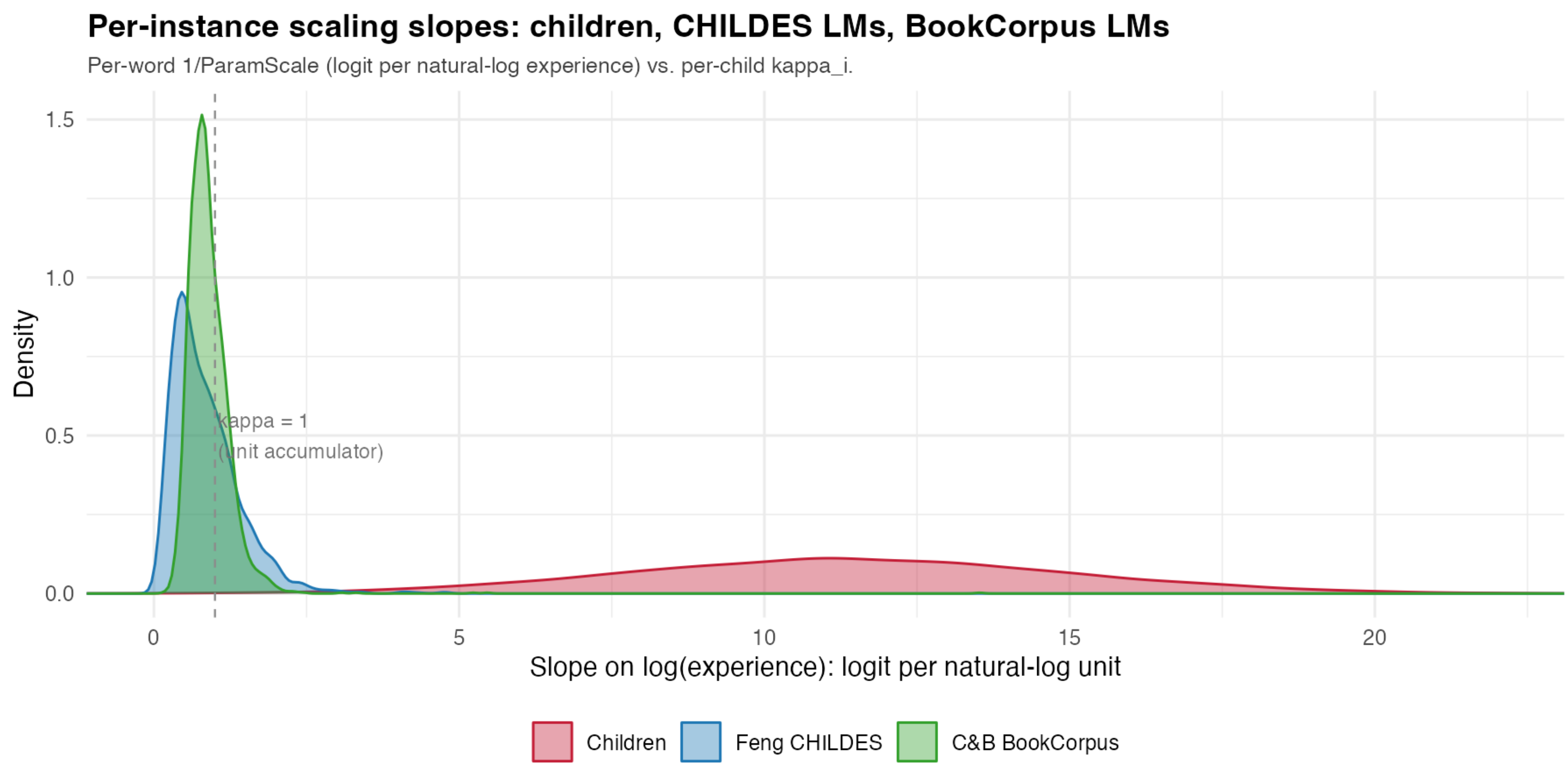
Apples-to-apples comparison

Test: retrain GPT-2 small on CHILDES, refit C&B's per-word sigmoid.

Setup:

- GPT-2 small (124M params, 12L × 768d × 12h)
- CHILDES train split, ~ 24.5M tokens, 20 epochs (Feng et al. §B spec)
- 3 random seeds ({0, 42, 123}) as a between-instance variance probe
- 73 log-spaced eval checkpoints; per-word mean NLL over 50 occurrences in the CHILDES validation set
- C&B 4-parameter logistic fit per (LM, word); per-word slope = $0.434/\mathrm{ParamScale}_w$
- 609 / 611 C&B CDI words are single tokens in the GPT2-CHILDES BPE

No increases in efficiency for LLMs



Minimal variance for LLMs (preliminary!)

- Between architectures (Chang & Bergen 2022, BookCorpus). Four LMs — BERT, GPT-2, biLSTM, LSTM — give median per-word slopes of 0.76, 0.81, 0.87, 0.96. Between-architecture SD ≈ 0.09.
- Between seeds (CHILDES-trained GPT-2 small). Three seeds on the same CHILDES data give median per-word slopes of 0.72, 0.74, 0.74. Between-seed SD ≈ 0.01.

Source	N	Median κ	between-LM SD
BookCorpus (C&B)	4	0.76–0.96	0.09
CHILDES seeds (this work)	3	0.72–0.74	0.01
Children	5000	10.3	3.5

Future work?

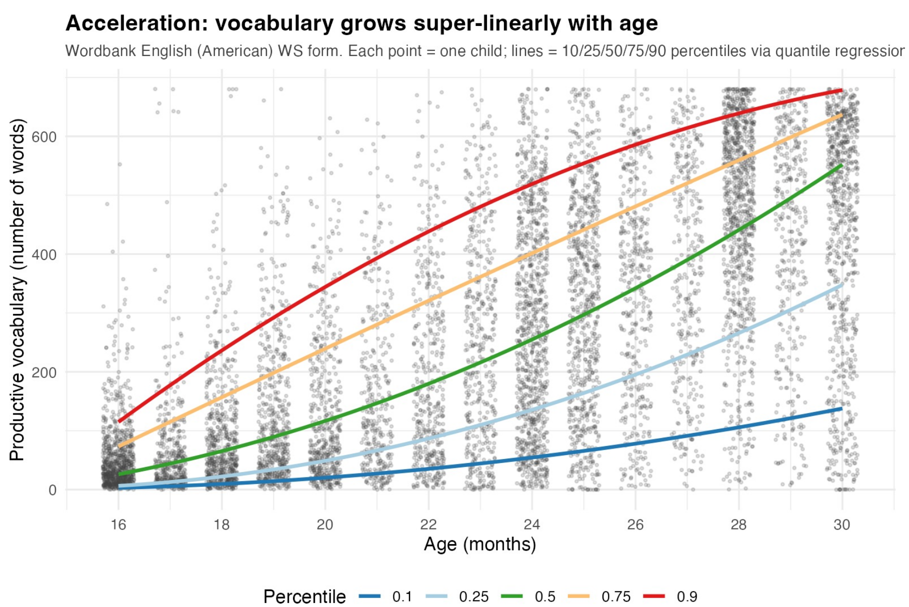
- Pilot to test for data-related variation. Train GPT-2 small on the two disjoint 10M-token chunks extractable from CHILDES, fix everything else.
- **Main study.** Three sources of between-LM variability, measured independently:
  – Seed variability.
  – Data variability — varied subsamples (CHILDES or TinyDialogues per pilot outcome), fixed seed and architecture.
  – Architecture variability — within-GPT-2 sweep over learning rate, width, depth.

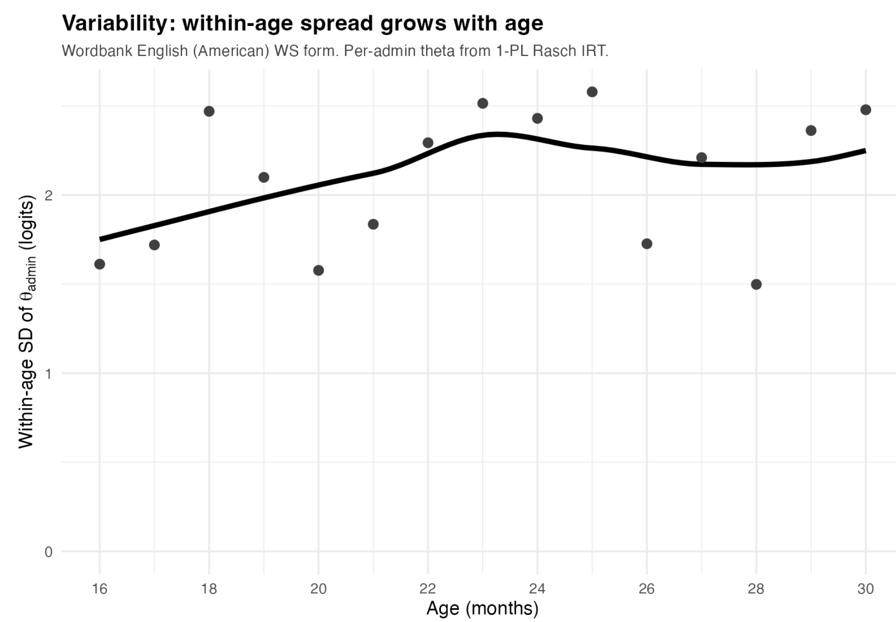
SYNTHESIS

Technical summary

- **Change in efficiency is real and population-level.** $\kappa_{\text{pop}} \approx 10$, confidently above the unit-accumulator boundary $\kappa_{\text{pop}} = 1$ (credit to Hidaka, 2013!).
- **Individual variability in efficiency is real too.** $\sigma_\zeta > 0$: children differ not just in baseline ability, but in *how their trajectories accelerate*.
- **Input quantity is a small share of between-child variance.** $\pi_\alpha \approx 0.91$ across 5 samples. Aligns with the meta-analytic $R^2 = 0.04$–0.07 of Coffey & Snedeker (2026).
- **LLMs are structurally different.** Same per-word sigmoid statistic gives $\kappa \sim 1$ for LMs vs. ~ 10 for children — a 10× gap with near-zero between-instance variance in LMs.

Two signatures, twin constraints on theory





Thanks!